HTTP and REST APIs

Methods, status codes and headers for designing and debugging web APIs

METHODS

`GET /orders`	read; safe and repeatable
`POST /orders`	create; not repeatable by default
`PUT /orders/42`	replace the whole resource
`PATCH /orders/42`	change some fields
`DELETE /orders/42`	remove; repeating it is harmless

SUCCESS AND REDIRECTS

`200 OK`	request worked, body has the result
`201 Created`	new resource; Location header points to it
`204 No Content`	worked, nothing to return
`301 / 308`	moved for good (308 keeps the method)
`302 / 307`	moved for now (307 keeps the method)
`304 Not Modified`	cached copy is still current

CLIENT ERRORS (4XX)

`400 Bad Request`	malformed or invalid input
`401 Unauthorized`	missing or bad credentials
`403 Forbidden`	known caller, not allowed
`404 Not Found`	no such resource
`409 Conflict`	clashes with current state
`422 Unprocessable Content`	well-formed but fails validation
`429 Too Many Requests`	slow down; see Retry-After

SERVER ERRORS (5XX)

`500 Internal Server Error`	unexpected failure on the server
`502 Bad Gateway`	an upstream server answered badly
`503 Service Unavailable`	overloaded or in maintenance
`504 Gateway Timeout`	an upstream server took too long

USEFUL HEADERS

`Content-Type: application/json`	format of the body sent
`Accept: application/json`	format the client wants back
`Authorization: Bearer <token>`	credentials for the request
`Cache-Control: no-store`	do not keep a copy
`ETag / If-None-Match`	cheap check whether data changed
`Idempotency-Key: <uuid>`	make a retried POST safe (if supported)

REST DESIGN HABITS

`/customers/7/orders`	plural nouns, nesting for ownership
`?status=paid&limit=20`	filter and page with query params
`?sort=-created_at`	sort; a leading minus means descending
`{"error": {"code": "...", "message": "..."}}`	one consistent error shape
`/v1/...`	version the API before the first breaking change

GOOD HABITS

- Use HTTPS everywhere; never put secrets in URLs.
- Return the most specific status code that is true.
- Document every endpoint's success and error responses.

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